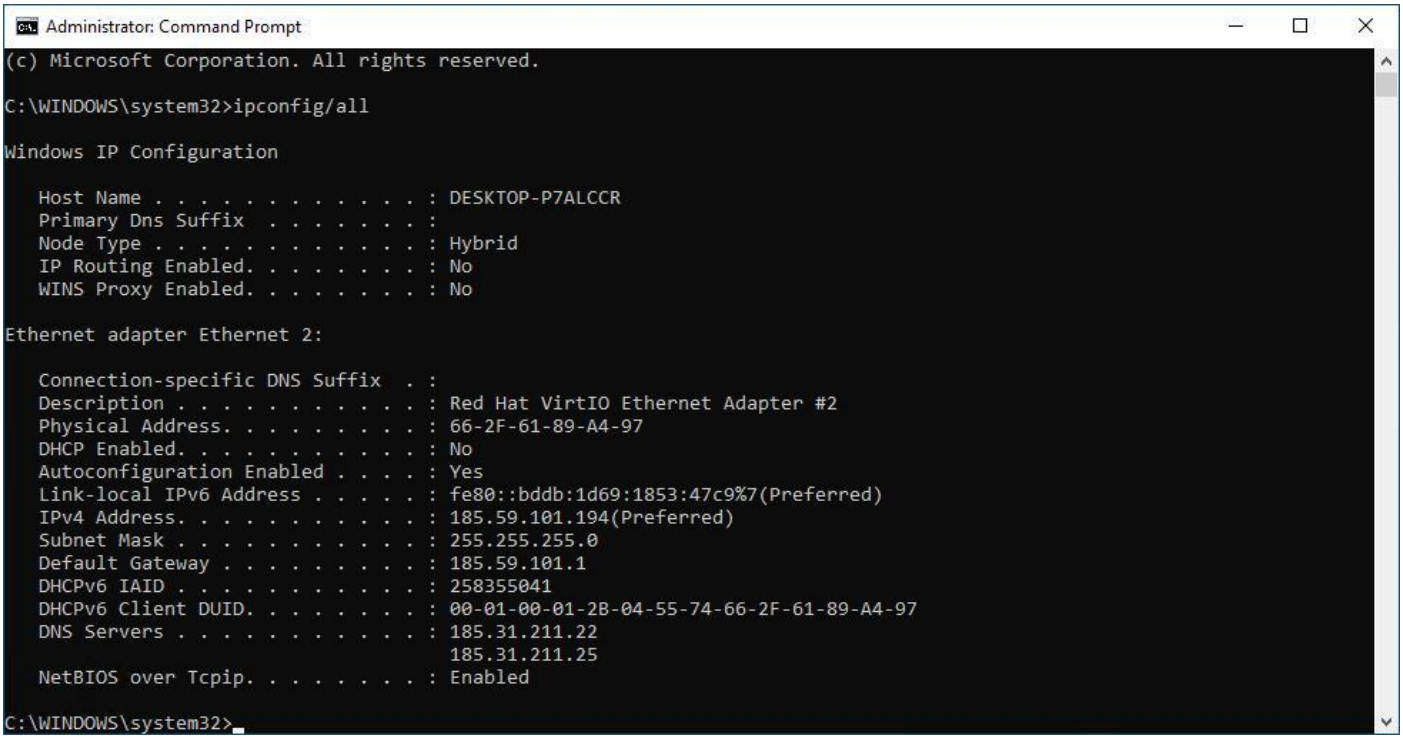
N+ Network Security

10/30/2026

Assignment-8

Prepare By
~pintu mer

1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.



2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Network Connectivity Test Using Ping

➔ First, I disconnected my Wi-Fi and ran the `ping www.google.com` command. The command failed because the internet connection was disconnected.

Then, I reconnected my Wi-Fi and ran the same command again. I received replies from Google's server.

Conclusion: Ping is used to check network connectivity. A successful reply means the destination is reachable, while a timeout or error indicates a connectivity or DNS problem.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Network Usage Monitoring

I opened Task Manager and played a YouTube video for 5 minutes. I monitored the network usage in the Performance tab.

- **Peak Network Usage:** 0.1Mbps
- **Highest Bandwidth Process:** Google Chrome / Microsoft Edge

Conclusion: Streaming a YouTube video increases network usage because video data is continuously downloaded from the internet.

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

Network Traffic Capture Using Wireshark

I used Wireshark to capture network traffic while browsing Flipkart/Instagram for 30 seconds. Then, I applied the http or tls display filter to view HTTP and HTTPS traffic.

Three types of information visible in filtered packets:

1. Source IP Address – Shows where the packet comes from.
2. Destination IP Address – Shows where the packet is going.
3. Protocol and Packet Length – Shows the protocol used and packet size.

Conclusion: Wireshark helps monitor and analyze network traffic. HTTPS traffic is encrypted to protect user data.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Troubleshooting Checklist for Slow Internet While Gaming

1. **Restart the Router:** Turn off the router for 30 seconds and turn it on again.
2. **Use Ethernet Cable:** Connect the gaming PC directly to the router using a LAN cable for a stable connection.
3. **Check Internet Speed:** Use an online speed test to check download and upload speed.

4. **Close Background Apps:** Close unnecessary applications and downloads that use internet bandwidth.
5. **Reduce Connected Devices:** Disconnect unused devices from the Wi-Fi network.
6. **Update Network Drivers:** Install the latest network adapter drivers.
7. **Move Closer to Router:** Keep the gaming device near the Wi-Fi router to improve signal strength.

My First Choice:

I would try using an Ethernet cable first because it provides a more stable connection and reduces Wi-Fi interference and latency.